

PES University — Department of Computer Science & Engineering

Lab 1: Requirements Engineering & UML Use-Case Modelling

Deliverable 1 — Requirements Table (Software Requirements Specification extract)

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Faculty	Dr Ashok Kumar Patil	Problem Statement	#65 — Sustainability & Green Tech: Urban Rooftop Gardening Co-op App

1. Scenario Under Analysis

A community urban-agriculture application for registering rooftop plots, logging crop yields and harvest schedules, sharing soil-maintenance tips, and trading surplus produce among co-op members. Target stakeholders / actors: Co-op Gardener and Community Leader.

2. Requirements Table — 5 Functional (FR-001 to FR-005) and 2 Non-Functional (NFR-001, NFR-002)

Req ID	Type	Description (“The system shall ...”)	Priority	Acceptance Criteria (measurable pass / fail)	Rationale
FR-001	Functional	The system shall allow a registered Co-op Gardener to log the seasonal planting date, crop type and plot ID for each crop, and shall automatically compute the estimated harvest window and the expected yield weight in kilograms.	High	Pass: For a crop logged with a valid planting date, the system displays a harvest window and an expected yield in kg, and raises a harvest alert within that predicted window. Fail: A planting-date sequence in which the planting date is later than the computed harvest date is accepted.	Core scheduling function of the co-op; harvest planning drives every downstream activity - trade listings, soil rotation and volunteer rostering.
FR-002	Functional	The system shall allow a Co-op Gardener to register a rooftop plot by capturing building address, plot area in square metres, daily sunlight hours and soil type, and shall set the plot status to “Active” only after a Community Leader approves the registration.	High	Pass: A submitted plot appears in the Community Leader’s approval queue within 5 s, and its status changes from “Pending” to “Active” on approval. Fail: A plot with a blank address or an area ≤ 0 m² is saved, or crops can be logged against a plot that is still “Pending”.	Verified plot data prevents duplicate or unsafe rooftop claims, and the approved plot record is the anchor for all yield and trade data.
FR-003	Functional	The system shall allow a Co-op Gardener to publish a surplus-produce listing stating crop, quantity in kilograms, availability window and exchange mode (sale price or barter), and shall display the listing on the community trade board.	High	Pass: A published listing becomes visible on the trade board of every community member showing the correct crop, quantity and exchange mode. Fail: A listing with a quantity ≤ 0 kg, or with an availability end-date earlier than its start-date, is published.	Surplus trading is the sustainability goal of the co-op - it converts excess yield into community value instead of food waste.
FR-004	Functional	The system shall allow a Co-op Gardener to reserve a surplus listing and settle the trade through an authorised payment or a barter confirmation, after which the listed quantity is decremented and a trade receipt is issued to both parties.	High	Pass: On successful payment authorisation the listing quantity decreases by exactly the reserved amount and both gardeners receive a receipt carrying the trade ID within 10 s. Fail: The same kilograms are reserved by two gardeners concurrently, or the quantity is decremented when the payment is declined.	Closes the trade loop and guarantees stock consistency; without settlement, listings cannot be audited or trusted by members.
FR-005	Functional	The system shall allow a Co-op Gardener to post soil-maintenance tips tagged by soil type and crop, allow any member to search and filter tips by those tags, and require Community Leader moderation before a tip is published.	Medium	Pass: A tip approved by the Community Leader and tagged “loamy, tomato” is returned by a search on either tag within 3 s. Fail: An unmoderated tip becomes publicly visible, or a search on an existing tag returns no matching approved tip.	Shared soil knowledge raises yields across the whole co-op and is the community-learning objective of the application.
NFR-001	Nonfunctional (Performance & Security)	The system shall update the produce trade board in real time for all community members, propagating any listing change within 2 seconds under a peak load of 500 concurrent users, with all client-server traffic carried over TLS 1.3.	High	Pass: Benchmarking under a simulated 500-concurrent-user peak load shows a 95th-percentile listing-refresh latency ≤ 2 s and a TLS 1.3 handshake on 100 % of sessions. Fail: 95th-percentile latency > 2 s, or any application endpoint is reachable over plain HTTP.	Stale listings cause double-booked produce and spoilage; trade traffic carries payment data, so transport security is mandatory.
NFR-002	Nonfunctional (Usability & Accessibility)	The system shall enable a first-time Co-op Gardener to complete rooftop-plot registration unaided in under 3 minutes, and all its screens shall conform to WCAG 2.1 Level AA.	Medium	Pass: In usability testing at least 8 of 10 first-time users complete registration in under 3 minutes without assistance, and an automated WCAG 2.1 audit reports zero Level A/AA violations. Fail: Median completion time ≥ 3 minutes, or any Level AA violation is reported.	Members are volunteer gardeners of mixed technical ability and age; poor usability or accessibility directly reduces co-op participation.

Priority key: High = required for the first release; Medium = required, may follow the first release. Shaded rows are the two non-functional requirements.

3. Peer Critique & Revision Notes

Requirement tables were exchanged with a peer and reviewed for clarity (is the “shall” statement unambiguous?), testability (does the acceptance criterion give a clear pass/fail?) and relevance to the scenario. The comments received and the revisions made are recorded below.

Req ID	Peer comment received and revision applied
FR-001	Peer comment: “State the unit of the expected yield.” Revision: description now specifies the expected yield weight in kilograms and the acceptance criterion tests an invalid planting-date sequence.
FR-004	Peer comment: “What happens if two members reserve the same produce at once?” Revision: a concurrent double-reservation was added as an explicit fail condition, and Alternate Flow 4a of the use-case specification covers it.
NFR-001	Peer comment: “‘Real-time’ is not testable - name the latency, the load and the security standard.” Revision: quantified as ≤ 2 s at the 95th percentile under 500 concurrent users over TLS 1.3.
NFR-002	Peer comment: “Add a measurable accessibility target, not just ‘easy to use’.” Revision: bound to WCAG 2.1 Level AA with an 8-of-10 unaided task-completion threshold.

4. Actors Extracted from the Approved Requirements

Actor	Role and purpose in interacting with the system
Co-op Gardener	Primary actor. Registers rooftop plots, logs planting and harvest data, posts and reads soil tips, lists surplus produce and trades it with other members.
Community Leader	Primary actor. Approves rooftop-plot registrations, moderates soil-maintenance tips and oversees community activity.
Payment Gateway	Secondary (external system) actor. Authorises and settles payments for surplus-produce trades.
Weather & Soil Data Service	Secondary (external system) actor. Supplies weather and soil data used to compute harvest windows and to raise crop-care alerts.

5. Use Cases Extracted from the Approved Requirements

UC ID	Use Case	Actor(s)	Traces to	Relationship
UC-01	Register Rooftop Plot	Co-op Gardener	FR-002	«include» UC-07 Verify Member Credentials
UC-02	Log Crop Planting & Harvest Schedule	Co-op Gardener	FR-001	«include» UC-08 Compute Harvest Window; extended by UC-09
UC-03	Share Soil Maintenance Tip	Co-op Gardener, Community Leader	FR-005	–
UC-04	List Surplus Produce for Trade	Co-op Gardener	FR-003	–
UC-05	Trade Surplus Produce	Co-op Gardener	FR-004	«include» UC-10 Process Trade Payment; extended by UC-11
UC-06	Approve Plot & Moderate Tips	Community Leader	FR-002, FR-005	–
UC-07	Verify Member Credentials	– (included)	FR-002	Included by UC-01
UC-08	Compute Harvest Window	Weather & Soil Data Service	FR-001	Included by UC-02
UC-09	Send Weather-Based Care Alert	Weather & Soil Data Service	FR-001	«extend» UC-02 (only on adverse-weather forecast)
UC-10	Process Trade Payment	Payment Gateway	FR-004	Included by UC-05
UC-11	Apply Co-op Member Discount	– (extending)	FR-004	«extend» UC-05 (only when the buyer is an eligible member)

UC-01 to UC-06 are the primary use cases invoked directly by an actor. The shaded rows UC-07 to UC-11 are the included and extending use cases that realise the «include» and «extend» relationships shown in Deliverable 2.